

Programme	B. Sc. Mathematics Honours			
Course Code	MAT5EJ305			
Course Title	HIGHER ALGEBRA			
Type of Course	Elective			
Semester	V			
Academic Level	300 - 399			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Fundamental Mathematics Concepts: Set, Functions, Logic			
Course Summary	This course explores topics that follow as a direct continuation of high school algebra, like the general theory of equations, and classification of second-degree curves and surfaces.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand and apply the algebraic methods used in solving polynomial equations of low degrees and place them in a general context	Ap	P	Internal Exam/Assignment/Seminar/Viva / End Sem Exam
CO2	Understanding of the fundamental concepts of algebraic equations, including the Identity Theorem and the Fundamental Theorem of Algebra.	U	C	Internal Exam/Assignment/Seminar/Viva / End Sem Exam
CO3	Analyse and evaluate various solutions of equations, including Cardan's Formulas and trigonometric solutions, and identify the irreducible cases.	An	C	Internal Exam/Assignment/Seminar/Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text	1. Geometry(2/e), David A Brannan, Mathew F. Esplen, Jeremy J Gray, Cambridge University Press (2012) ISBN: 978-1-107-64783-1 2. Theory of Equations, J. V. Uspensky, McGraw Hill (1948), ISBN:07-066735-7			
Module	Unit	Content	Hrs (48+12)	Ext. Marks (70)
I	Theory of Equations		16	
	1	Chapter II -Section 3: Division of Polynomials		
	2	Chapter II -Section 4: The Remainder Theorem		
	3	Chapter II- Section 5: Synthetic Division		
	4	Chapter II- Section 7: Taylor’s Formula		
	5	Chapter III - Section 1: Algebraic Equations		
	6	Chapter III - Section 2: Identity Theorem		
	7	Chapter III - Section 3: The Fundamental Theorem of Algebra		
II	Cubic And Biquadratic Equations		16	
	8	Chapter III - Section 4: Imaginary Roots of Equations with Real Coefficients		
	9	Chapter III - Section 5: Relations Between Roots and Coefficients		
	10	Chapter IV - Section 1: Limits of Roots Section 2: A Method to Find an Upper Limit of Positive Roots		
	11	Chapter IV - Section 3: Limit for Moduli of Roots		
	12	Chapter V - Section 1: What is the “Solution” of an Equation?, Section 2: Cardan’s Formulas, Section 3: Discussion of Solution		
	13	Chapter V - Section 4: Irreducible Case Section 5: Trigonometric Solution		
	14	Chapter V- Section 6: Solution of Biquadratic Equations		

III	Conic Sections		12
	15	Section 1.1.1: Conic Sections, Section 1.1.2: Circles	
	16	Section 1.1.3: Focus-Directrix Definition of the Non-Degenerate Conics	
	17	Section 1.1.4: Focal Distance Properties of Ellipse and Hyperbola	
	18	Section 1.1.5: Dandelin Spheres	
IV	Quadric Surfaces		4
	19	Section 1.2.2: Reflections	
	20	Section 1.3: Recognizing Conics	
	21	Section 1.4.1: Quadric Surfaces in $\mathbb{R}^3$	
	22	Section 1.4.2: Recognizing Quadric Surfaces	
V	Open Ended Module: Affine Maps		12
	1	Geometry and Transformations - What is Euclidean Geometry? Isometry, Euclidean properties, Euclidean transformation, Euclidean-Congruence	
	2	Affine Transformations, Basic Properties of Affine Transformations	
	3	Fundamental Theorem of Affine Geometry	

References:

- Higher Algebra, Barnard & Child, St. Martin’s Press, NY, USA (Public Domain, Copyright exhausted)
- Thomas & Finney, Calculus & Analytic Geometry, Addison Wesley
- George A Jennings: Modern Geometry with Applications Universitext, Springer (1994) ISBN: 0-387-94222-X
- Walter Meyer: Geometry and its Application(2/e) Elsever, Academic Press(2006) ISBN: 0-12-369427-0

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	3	3	2	3	1	2	1	3	0	1
CO 2	3	3	2	2	3	1	2	1	3	0	1
CO 3	2	3	3	2	3	1	3	1	3	0	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓